

A First Course in Real Analysis by Berberian

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1 Chapter Name

1.1 Axioms for $\mathbb{R}$

Problem 1

1. If $\rho : F \rightarrow G$ is an isomorphism of fields, then the inverse mapping $\rho^{-1} : G \rightarrow F$ is also an isomorphism. Thus $F \cong G \implies G \cong F$.
2. If $\rho : F \rightarrow G$ is a monomorphism of fields, then $\rho(1) = 1$.
3. The field $\mathbb{Q}$ of rational numbers is not isomorphic to the field $\mathbb{Q}(t)$ of rational forms over $\mathbb{Q}$.

Proof. Suppose $\rho : F \rightarrow G$ is an isomorphism of fields. Let $a, b \in G$. Since ρ is bijective, there exists unique $x, y \in F$ such that $\rho(x) = a, \rho(y) = b$, so $\rho^{-1}(a) = x, \rho^{-1}(b) = y$. Now

$$\rho^{-1}(a + b) = \rho^{-1}(\rho(x) + \rho(y)) = (\rho^{-1} \circ \rho)(x + y) = x + y = \rho^{-1}(a) + \rho^{-1}(b).$$

Similarly,

$$\rho^{-1}(ab) = \rho^{-1}(\rho(x)\rho(y)) = (\rho^{-1} \circ \rho)(xy) = xy = \rho^{-1}(a)\rho^{-1}(b).$$

Thus ρ^{-1} is a homomorphism, and it is bijective since ρ is, so ρ^{-1} is an isomorphism. ■

Proof. Suppose $\rho : F \rightarrow G$ is a monomorphism of fields. Since ρ is injective and $\rho(0) = 0$, we have $\rho(1) \neq 0$. Then

$$1^2 = 1 \iff \rho(1^2) = \rho(1) \iff \rho(1)\rho(1) = \rho(1).$$

Cancelling $\rho(1) \neq 0$ gives $\rho(1) = 1$. ■

Proof. Suppose $\rho : \mathbb{Q} \rightarrow \mathbb{Q}(t)$ is a monomorphism. By (ii) we have $\rho(1) = 1$, then $\rho(n) = n$ for all $n \in \mathbb{Z}_{>0}$ by induction. Since $\rho(-1) + \rho(1) = \rho(0) = 0$, we have $\rho(-1) = -1$, so $\rho(n) = n$ for all $n \in \mathbb{Z}$. For $p/q \in \mathbb{Q}$, we have $q \cdot (p/q) = p$, so $\rho(q)\rho(p/q) = \rho(p)$, giving $\rho(p/q) = p/q$. Thus $\rho(x) = x$ for all $x \in \mathbb{Q}$, so ρ is not surjective, a contradiction. ■

Problem 2

In the field $F = \mathbb{Q} + i\mathbb{Q}$ of Gaussian rationals, if $\alpha = r + is$ define $\bar{\alpha} = r - is$. Show that the mapping $\rho : F \rightarrow F$ defined $\rho(\alpha) = \bar{\alpha}$ is an automorphism of F , and that, for every nonzero element α , $\alpha^{-1} = (r^2 + s^2)^{-1}\bar{\alpha}$.

Proof. Let $\alpha, \beta \in F$ and suppose $\alpha = a + ib, \beta = x + iy$. Then

$$\rho(\alpha + \beta) = \rho((a + x) + i(b + y)) = (a + x) - i(b + y) = a - ib + x - iy = \rho(\alpha) + \rho(\beta).$$

Similarly,

$$\rho(\alpha\beta) = \rho((a + ib)(x + iy)) = \rho((ax - by) + i(ay + xb)) = (ax - by) - i(ay + xb) = \rho(\alpha)\rho(\beta).$$

Since $\rho(\rho(\alpha)) = \bar{\bar{\alpha}} = \alpha$, ρ is its own inverse and thus bijective, so ρ is an automorphism. For the second part, we verify directly

$$\alpha \cdot (r^2 + s^2)^{-1}\bar{\alpha} = (r + is)(r - is)(r^2 + s^2)^{-1} = (r^2 + s^2)(r^2 + s^2)^{-1} = 1.$$

Thus $\alpha^{-1} = (r^2 + s^2)^{-1}\bar{\alpha}$. ■

1.2 The Order Axioms

Problem 1

In an ordered field, $0 < a < b \iff b^{-1} < a^{-1}$.

Proof. Suppose $0 < a < b$.

$$a^{-1} - b^{-1} = a^{-1}(b - a)b^{-1}.$$

On the rhs, $b - a > 0$ by hypothesis. Since $b > a > 0$, by Theorem 1.2.3 (12) we have $a^{-1} > 0$ and $b^{-1} > 0$. Then by O_2 the product on the rhs is in P , thus $b^{-1} < a^{-1}$.

Conversely, suppose $b^{-1} < a^{-1}$. If $a \leq 0$ then $a^{-1} \leq 0$, and if $b \leq 0$ then $b^{-1} \leq 0$, so $b^{-1} < a^{-1} \leq 0$, contradicting $b^{-1} < a^{-1}$. Thus $a > 0$ and $b > 0$. If $0 < b < a$, then by the forward direction $a^{-1} < b^{-1}$, contradicting $b^{-1} < a^{-1}$. Thus $0 < a < b$. ■

Problem 2

In an ordered field, $a^2 + b^2 = 0 \implies a = b = 0$. Similarly, for a sum of n squares ($n = 3, 4, 5, \dots$).

Proof. Suppose $a^2 + b^2 = 0$. Suppose for contradiction that $a \neq 0$ or $b \neq 0$, WLOG $a \neq 0$. Then $a^2 > 0$ by Theorem 1.2.3 (9) and $b^2 \geq 0$, so $a^2 + b^2 > 0$ by O_1 . This contradicts $a^2 + b^2 = 0$. Thus $a = b = 0$. For a sum of n squares $x_1^2 + \dots + x_n^2 \geq x_1^2 \geq 0$. But $x_1^2 + \dots + x_n^2 = 0$ thus $x_1 = 0$ and for each x_i it follows similarly. ■

Problem 3

In an ordered field, $a > b > c \geq 0$ and $a \leq b + c$, then

$$\frac{a}{1+a} \leq \frac{b}{1+b} + \frac{c}{1+c}.$$

Proof. Suppose $a > b > c \geq 0$ and $a \leq b + c$. Then

$$\begin{aligned} \frac{b}{1+b} + \frac{c}{1+c} - \frac{a}{1+a} &= \frac{b(1+a)(1+c) + c(1+b)(1+a) - a(1+b)(1+c)}{(1+a)(1+b)(1+c)} \\ &= \frac{b + ab + bc + abc + c + ac + bc + abc - a - ab - ac - abc}{(1+a)(1+b)(1+c)} \\ &= \frac{b + c - a + 2bc + abc}{(1+a)(1+b)(1+c)} \\ &= \frac{(b + c - a) + bc(2 + a)}{(1+a)(1+b)(1+c)}. \end{aligned}$$

Since $a \leq b + c$ we have $b + c - a \geq 0$, and since $a > b > c \geq 0$ all of b, c, a are non-negative so $bc(2 + a) \geq 0$. Thus the numerator is ≥ 0 and the denominator is > 0 , giving

$$\frac{b}{1+b} + \frac{c}{1+c} - \frac{a}{1+a} \geq 0. \quad \blacksquare$$

Problem 5

Let F be an ordered field, a and b any elements of F , and n an odd positive integer. Prove:

1. $a < b \iff a^n < b^n$;
2. $a = b \iff a^n = b^n$;
3. $a > b \iff a^n > b^n$;

Proof. (i) Suppose $a < b$. If $a \geq 0$ we have Theorem 1.2.8. Thus suppose $a < 0$. There are three cases. Suppose $b = 0$. Since n is odd $a^n < 0$ thus $a^n < b^n$. Suppose $b > 0$. Since n is odd $a^n < 0$ thus $a^n < b^n$. Suppose $b < 0$. We have $-a > -b$ and by Theorem 1.2.8 $(-a)^n > (-b)^n \iff (-1)^n a^n > (-1)^n b^n$. Since n is odd $(-1)^n = -1$ so we have $-(a^n) > -(b^n)$. Multiplying by -1 gives $a^n < b^n$. The converse follows by trichotomy. If $a^n < b^n$ then $a = b$ is impossible and $a > b$ is impossible so $a < b$.

(ii) is trivial. (iii) is (i) with the roles of a and b reversed. $\blacksquare$

Problem 6

Let $\mathbb{Q}(t)$ be the field of rational forms over $\mathbb{Q}$ and let S be the set of all nonzero elements $r(t) = p(t)/q(t)$ of $\mathbb{Q}(t)$ such that the leading coefficients of the polynomials p and q have the same sign, that is, such that $p(t) = a_0 + a_1 t + \dots + a_m t^m$, $q(t) = b_0 + b_1 t + \dots + b_n t^n$ with $a_m b_n > 0$.

1. $\mathbb{Q}(t)$ is an ordered field, with S as a set of positive elements.
2. For the ordering defined in (i), $n1 < t$ for every positive integer n .

Proof. Let $f, g \in \mathbb{Q}(t)$. Write $f = \frac{p_1}{q_1}$, $g = \frac{p_2}{q_2}$ where $p_1, p_2, q_1, q_2 \in \mathbb{Q}[t]$, with leading coefficients a_m, c_k of p_1, p_2 and b_n, d_ℓ of q_1, q_2 respectively.

(i) We verify the three axioms for S to be a set of positive elements.

Trichotomy. Let $f \in \mathbb{Q}(t)$, $f \neq 0$. Then $f = p_1/q_1$ with $p_1 \neq 0$, so $a_m b_n \neq 0$. Either $a_m b_n > 0$, in which case $f \in S$, or $a_m b_n < 0$, in which case $(-a_m) b_n > 0$ so $-f \in S$. These are mutually exclusive since $a_m b_n > 0$ and $a_m b_n < 0$ cannot both hold.

Closure under addition. Suppose $f, g \in S$, so $a_m b_n > 0$ and $c_k d_\ell > 0$. Then

$$f + g = \frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{p_1 q_2 + p_2 q_1}{q_1 q_2}.$$

The leading coefficient of $q_1 q_2$ is $b_n d_\ell$. The leading coefficient of $p_1 q_2 + p_2 q_1$ is determined by the highest-degree term; since $\deg(p_1 q_2) = m + \ell$ and $\deg(p_2 q_1) = k + n$, if $m + \ell \neq k + n$ the leading coefficient is $a_m d_\ell$ or $c_k b_n$ respectively,

both of which are positive (as products of same-sign pairs); if $m + \ell = k + n$ the leading coefficient is $a_m d_\ell + c_k b_n > 0$. In all cases the numerator's leading coefficient times $b_n d_\ell > 0$, so $f + g \in S$.

Closure under multiplication. Suppose $f, g \in S$, so $a_m b_n > 0$ and $c_k d_\ell > 0$. Then

$$fg = \frac{p_1 p_2}{q_1 q_2}.$$

The leading coefficient of $p_1 p_2$ is $a_m c_k$ and of $q_1 q_2$ is $b_n d_\ell$. Then $(a_m c_k)(b_n d_\ell) = (a_m b_n)(c_k d_\ell) > 0$, so $fg \in S$.

Thus S makes $\mathbb{Q}(t)$ an ordered field.

(ii) Fix a positive integer n . We must show $t - n \cdot 1 \in S$, i.e. $t - n \in S$ (identifying $n \cdot 1$ with the constant polynomial n). We have $t - n = \frac{t-n}{1}$. The leading coefficient of the numerator $t - n$ is $1 > 0$ and the leading coefficient of the denominator 1 is $1 > 0$, so $(1)(1) > 0$ and thus $t - n \in S$, giving $n < t$. ■

Problem 10

If a, b are elements of an ordered field, such that $a \leq b + c$ for every $c > 0$, then $a \leq b$.

Proof. Suppose a, b are elements of an ordered field and $a \leq b + c$ for every $c > 0$. For contradiction, suppose $a > b$. Then, by Exercise 2, we have, $b < \frac{a+b}{2} < a$. Then, notice $(a+b)/2 - b = (a-b)/2$ and, since $a - b > 0$, $(a+b)/2 - b > 0$. Then, letting $c = (a+b)/2 - b > 0$, the hypothesis gives $a \leq b + c = (a+b)/2$, contradicting $(a+b)/2 < a$. ■

Problem 13

(Bernoulli's inequality) In an ordered field, let x be any element such that $x \geq -1$. Prove that

$$(1+x)^n \geq 1+nx$$

for every positive integer n . If $x > 1$ and $x \neq 0$, then $(1+x)^n > 1+nx$ for all $n \geq 2$.

Proof. Let $x \geq -1$, so that $1+x \geq 0$. For the base case we have $(1+x)^n = (1+x)^1 = 1+x$ and $1+nx = 1+(1)x = 1+x$ so $(1+x)^n - (1+nx) = (1+x) - (1+x) = 0$. So $(1+x)^n \geq 1+nx$ as required. Suppose the inequality holds for $n-1$ so that $(1+x)^{n-1} \geq 1+(n-1)x$. Since $1+x \geq 0$, multiplying both sides by $1+x$ preserves the inequality, giving $(1+x)^n = (1+x)(1+x)^{n-1} \geq (1+x)(1+(n-1)x) = 1+nx + (n-1)x^2 \geq 1+nx$, as required. ■

Problem 18

In an ordered field, let P be the set of elements > 0 . Prove that P has neither a smallest element nor a largest element.

Proof. Suppose $p \in P$ is the largest element. But then $p+1 \in P$ and $p+1-p = 1 > 0$ so $p+1 > p$, contradicting that p is the largest element. Suppose $p \in P$ is the smallest element. But, by Exercise 2, since $0 < p$ we have $0 < p/2 < p$, and $p/2 \in P$, contradicting that p is the smallest element. ■

Problem 20

In an ordered field,

$$\left(1 - \frac{1}{n^2}\right)^n > 1 - \frac{1}{n}$$

for every integer $n \geq 2$ (Exercise 13); infer that

$$\left(1 + \frac{1}{n-1}\right)^{n-1} < \left(1 + \frac{1}{n}\right)^n.$$

Proof. Factoring, $1 - \frac{1}{n^2} = \left(1 + \frac{1}{n}\right)\left(1 - \frac{1}{n}\right)$, so by Exercise 13,

$$\left(1 + \frac{1}{n}\right)^n \left(1 - \frac{1}{n}\right)^n > 1 - \frac{1}{n}.$$

Since $1 - \frac{1}{n} > 0$, divide by $1 - \frac{1}{n}$ to get $\left(1 + \frac{1}{n}\right)^n \left(1 - \frac{1}{n}\right)^{n-1} > 1$. Since $1 - \frac{1}{n} = \left(1 + \frac{1}{n-1}\right)^{-1}$, this becomes $\left(1 + \frac{1}{n}\right)^n \left(1 + \frac{1}{n-1}\right)^{-(n-1)} > 1$, so $\left(1 + \frac{1}{n-1}\right)^{n-1} < \left(1 + \frac{1}{n}\right)^n$. ■

1.3 Bounded Sets, LUB and GLB

Problem 1

Let F be an ordered field, $a \in F$. The subset

$$\{x \in F : x < a\}$$

is bounded above, but not below; the subset

$$\{x \in F : x > a\}$$

is bounded below, but not above. Each of the intervals of 1.3.2 is the intersection of two unbounded sets; for example,

$$[a, b) = \{x : x \geq 0\} \cap \{x : x < b\}.$$

Solution: Alrighty Then.

Problem 2

On the field $\mathbb{Q}$ of rational numbers, the subset $\mathbb{P} = \{1, 2, 3, \dots\}$ is not bounded above.

Proof. Suppose there is an upper bound $p \in \mathbb{P}$. Then $(p+1) - p = 1 > 0$ so $p+1 > p$, contradicting that p is an upper bound, since $p+1 \in \mathbb{P}$. ■

Problem 3

In the ordered field $\mathbb{Q}(t)$ of 1.2, Exercise 6, the subset $\mathbb{P} = \{1, 2, 3, \dots\}$ is bounded; it has no least upper bound.

Proof. By Exercise 6(ii), $n \cdot 1 < t$ for every positive integer n , so t is an upper bound for $\mathbb{P}$ and $\mathbb{P}$ is bounded. Suppose, for contradiction, that $u \in \mathbb{Q}(t)$ is a least upper bound for $\mathbb{P}$. Then by Problem 3, $\mathbb{P}$ has no least upper bound. ■

Problem 4

If, in an ordered field, $m = \inf A$ and $m' > m$, show that there exists an element $a \in A$ such that $m \leq a < m'$.

Proof. Since $m = \inf A$ we know $A \neq \emptyset$. Since $m' > m$ and m is the greatest lower bound of A , m' cannot be a lower bound of A . So there exists $a \in A$ with $a < m'$. But m is a lower bound of A , so $m \leq a$. Thus $m \leq a < m'$, as required. ■

1.4 The Completeness Axiom

Problem 1

Let A and B be nonempty subsets of $\mathbb{R}$ that are bounded above, and write

$$A + B = \{a + b : a \in A, b \in B\}.$$

Prove that $A + B$ is bounded above and that $\sup(A + B) = \sup(A) + \sup(B)$.

Proof. Since A and B are bounded above $\sup A$ and $\sup B$ exists. Let $x = a + b \in A + B$. Then $(a + b) - (\sup A + \sup B) = (a - \sup A) + (b - \sup B)$. Since $a \leq \sup A$ and $b \leq \sup B$, we have $a + b \leq \sup A + \sup B$ for all $x \in A + B$. That is, $A + B$ is bounded above.

Let $\alpha = \sup A$, $\beta = \sup B$ and notice $a \leq \alpha$ and $b \leq \beta$ for all $a \in A$ and $b \in B$ so $a + b \leq \alpha + \beta$. It suffices to show that if c is any upper bound for $A + B$ then $\alpha + \beta \leq c$. If $a \in A$ then $a + b \leq c$ for all $b \in B$ so $b \leq c - a$ is an upper bound of B . It follows that $\beta \leq c - a$, thus $a \leq c - \beta$ for all a . Then $\alpha \leq c - \beta$ so $\alpha + \beta \leq c$ as required. ■

Problem 2

Let A and B be nonempty sets of positive real numbers that are bounded above, and write

$$AB = \{ab : a \in A, b \in B\}.$$

Prove that AB is bounded above and that $\sup(AB) = (\sup A)(\sup B)$.

Proof. Since A and B are bounded above $\sup A$ and $\sup B$ exists. Let $\alpha = \sup A$, $\beta = \sup B$ and notice $a \leq \alpha$ and $b \leq \beta$ for all $a \in A$ and $b \in B$, so $ab \leq \alpha\beta$ since $a, b > 0$. That is, AB is bounded above.

It suffices to show that if c is any upper bound for AB then $\alpha\beta \leq c$. If $a \in A$ then $ab \leq c$ for all $b \in B$, so since $a > 0$, $b \leq c/a$ is an upper bound of B . It follows that $\beta \leq c/a$, thus $a \leq c/\beta$ for all a . Then $\alpha \leq c/\beta$ so $\alpha\beta \leq c$ as required. ■

Problem 5

(Principle of Mathematical Induction) If T is a subset of $\mathcal{P}$ such that $1 \in T$ and such that $n \in T \implies n + 1 \in T$, then $T = \mathcal{P}$.

Proof. Suppose $T \neq \mathcal{P}$, and let $m = \min(\mathcal{P} - T)$. Since $1 \in T$, $m \neq 1$, so $m - 1 \in \mathcal{P}$, and since m is the smallest element of $\mathcal{P} - T$, $m - 1 \in T$. Clearly $(m - 1) + 1 = m \in T$, a contradiction since $m \in \mathcal{P} - T$. ■